AI acknowledgement coversheet

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Your details

Name: Jie Wen

Student Number: 48559489

Assessment details

Course: DECO7230

Assignment: Design Evaluation 1

AI use

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- AI tool (e.g. Copilot, ChatGPT, Claude, Gemini)
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- prompt(s) used
- relevant part, section, or page of the assessment
- date.

Complete the table below with this information.

Tool	Use	Prompt(s)	Section	Date
Chatgpt5o	Report structure and written content creation	Based on the testing plan and usability test results, draft a Design Evaluation 1 Report	Objective and Validation Metrics; Method; Results; Reflection	31 August
Gemini	Used to modify the written tone	Rewrite this paragraph so that the tone is appropriate for a university assignment	Entire work	1 September

This coversheet has been adapted with thanks to BIOM3020 Integrated Endocrinology.

Objective and Validation Metrics

Concept under evaluation. IdeaSpace XR reimagines a Miro-like brainstorming workspace as an immersive spatial environment in which ideas are represented as moveable cards and organised across labelled boards. Prototype 1 was a horizontal Unity prototype focused on the higher-level user goal of organising ideas, rather than treating select, move and place as separate tasks.

Aim	What was tested	Operational success criterion	Evidence collected
A1 Spatial comprehension	Whether users understand the role of Unsorted and category boards.	Participant can identify the intended spatial organisation logic and explain where an idea belongs without corrective explanation.	Observed interpretation, hesitation, post-test Q1-Q3.
A2 Interaction discoverability	Whether a new user can work out how to move a card.	Participant completes the card move without evaluator assistance or repeated failed attempts.	Task completion, help, time, hesitation, post-test Q4-Q5.
A3 Concept comprehension	Whether the interaction communicates the purpose of IdeaSpace XR.	Participant can explain the overall idea as organising/categorising ideas after the task.	Task 3 explanation and overall rating.

Method

Five peer participants (P1-P5) completed a short task-based usability test in a controlled setting. Think-aloud had been planned, but participants preferred not to verbalise their actions, so the actual session relied on direct observation and a five-question post-test interview. Participants first interpreted the spatial zones, then moved a problem card from Unsorted to Problems. Recorded measures included completion, time, assistance, hesitation/errors, purpose comprehension and a 1-5 overall assessment.

Result

Observed result	Raw outcome
Core card movement	5/5 completed successfully.
Assistance	4/5 completed without help; P5 required one gentle nudge.
Completion time	Mean 2:07; median 2:14; range 1:00-3:15.
Initial hesitation	4/5 showed a pause, uncertainty or hesitation before/around the first interaction.
Zone/label interpretation	P1, P2 and P4 were unambiguous; P3 questioned the meaning of “Problems”; P5 understood “Unsorted” but did not immediately realise the cards were interactive.
Overall purpose	All 5 participants could explain the organising/categorising purpose after completing the interaction; P5 reached this understanding after the prompt.
Overall assessment	2 participants rated 5/5 and 3 rated 4/5; mean = 4.4/5.
Post-test comments	P1 requested visible user hands; P2 requested multi-select and noted overlap between “Positive Finding” and “Opportunity”; P3 requested category descriptions/examples; P4 and P5 requested interaction highlighting/prompts.

Analysis and Insights

Insight 1 - The spatial metaphor is understandable, but category semantics are not fully self-explanatory.
The strongest pattern is that participants generally understood the board-based spatial organisation, yet three participants raised semantic issues: P1 was sometimes unsure what an idea meant, P2 saw overlap between “Positive Finding” and “Opportunity”, and P3 interpreted “Problems” as possibly meaning unfinished tasks. The structure therefore works better than the vocabulary. The design needs to reduce interpretation effort at the category level rather than redesigning the entire spatial layout.

Insight 2 - The main usability bottleneck is discovering how to interact, not deciding where to place a card.
All participants completed the move, but 4/5 showed some first-interaction hesitation and P5 needed a hint. P5 explicitly said the categories were understandable while moving the card was harder; P4 and P5 independently suggested highlights/prompts. This indicates an affordance and feedforward problem: the card looks understandable as content, but not sufficiently actionable as an interactive object.

Insight 3 - Users expect stronger embodied and system feedback in an XR concept.
The earlier paper-prototype test already raised questions about whether cards would snap into place and whether successful categorisation would provide visual feedback. The Unity test reinforced this expectation: P1 wanted visible user hands and P4/P5 wanted interaction highlighting. The repeated request across prototype stages suggests that visible hands/controller cues, hover feedback and placement confirmation are part of the interaction model, not cosmetic polish.

Insight 4 - A one-card workflow demonstrates feasibility but not scalable organisation.
P2 requested multi-selection because moving notes one-by-one becomes time-consuming. Since the recorded task used one representative card, the current data support basic usability but cannot validate efficiency for organising many ideas. Scalability remains a design question for the next prototype.

Evaluation of Aims

Aim	Evaluation	Why	Implication
A1 Spatial comprehension	**PARTIALLY VALIDATED**	The spatial board logic was broadly understood, but category labels created ambiguity for P1-P3.	Keep the spatial-board structure; clarify category meaning.
A2 Interaction discoverability	**NOT YET VALIDATED**	Although 5/5 succeeded, 4/5 hesitated and P5 required assistance.	Add explicit feedforward and proximity feedback before retesting.
A3 Concept comprehension	**VALIDATED**	All participants could explain the organising/categorising purpose after the task.	Retain spatial organisation as the concept’s core workflow.

Overall concept evaluation: Partially confirmed. Prototype 1 validates the high-level idea of spatially organising cards, but it does not yet validate that the interaction is immediately discoverable, more efficient than a 2D workflow, or suitable for collaboration.

Concept Iteration

The iteration below keeps what the test supported and changes only elements for which there is direct evidence. This creates a traceable chain from result → insight → design change.

Evidence	Design issue	Iteration	Reason / expected effect
P5 needed a hint; P4/P5 requested highlights/prompts.	Weak interaction discoverability.	Add proximity highlight plus an in-world grab cue (e.g., “Press E to pick up” in desktop testing; controller/hand cue in XR).	Makes the first possible action visible before the user has to guess.
Paper test asked about snap/confirmation; P4/P5 wanted feedback.	Unclear placement state.	Add snap-to-board behaviour, hover state and brief confirmation after placement.	Creates a clear action-response loop and reduces uncertainty about successful categorisation.
P1-P3 reported label/meaning ambiguity.	Category semantics overlap.	Add a one-line descriptor/example under each board title and stronger visual differentiation between categories.	Preserves the spatial model while reducing interpretation effort.
P1 requested visible hands.	Prototype feels less embodied than the XR concept promises.	Use visible virtual hands/controllers and direct grab/release interaction in the next vertical prototype.	Aligns the interaction with the intended immersive, embodied concept.
P2 requested multi-select.	One-by-one organisation may not scale.	Prototype grouped selection / multi-card movement for a later test.	Tests whether spatial organisation remains efficient with realistic numbers of ideas.

IdeaSpace XR is an immersive brainstorming workspace where ideas exist as graspable spatial cards rather than items on a flat canvas. Users create and organise ideas across clearly explained spatial boards, with immediate feedforward and feedback through proximity highlights, grab cues and snap-to-board confirmation. Future iterations extend the validated organising metaphor into direct hand-based manipulation, grouped movement and collaborative idea refinement.

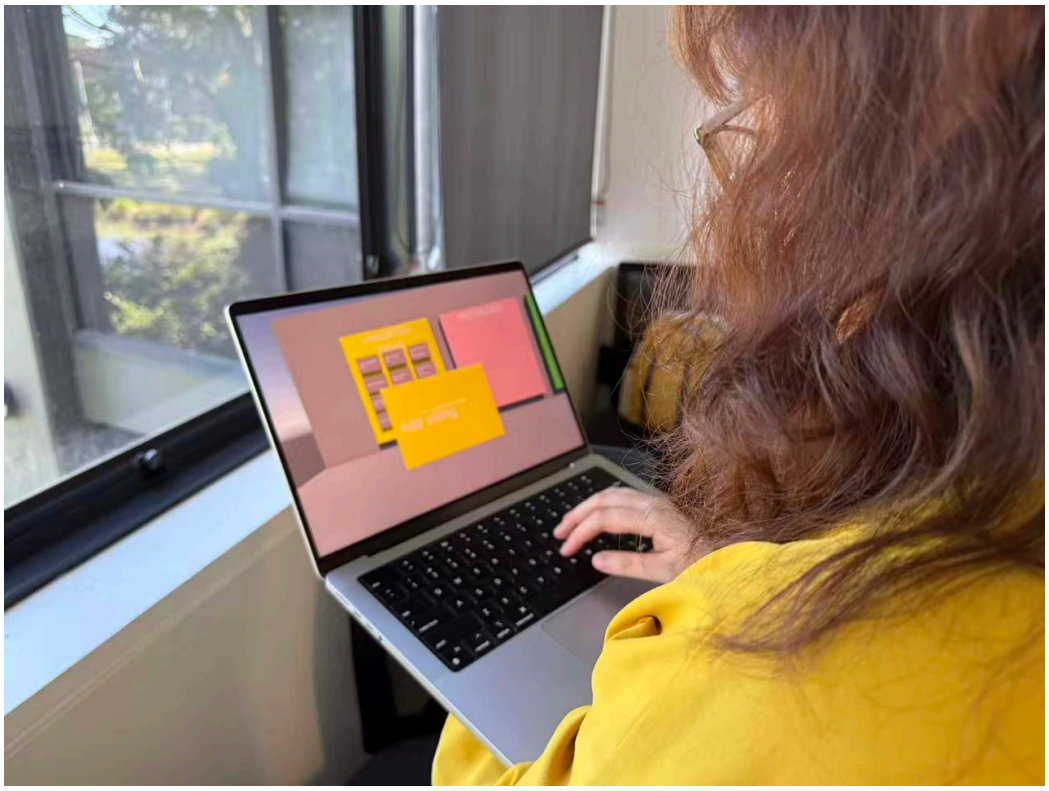
Reflection and Next Steps

The combination of observation, task completion data and short post-test questions was still effective, but not using think-aloud reduced access to participants' reasoning during the interaction. I therefore relied more on visible hesitation, task behaviour and post-test explanations, and avoided inferring motives that were not directly supported. The session also showed why completion rate alone is insufficient: a 100% completion rate would have hidden the discoverability problem revealed by hesitation and assistance. However, the study used only five peer participants in a controlled Unity setup, the recorded workflow centred on one card, and one participant received a prompt. The test therefore cannot support claims that IdeaSpace XR is faster or more intuitive than Miro, that multi-card organisation is efficient, or that collaboration works.

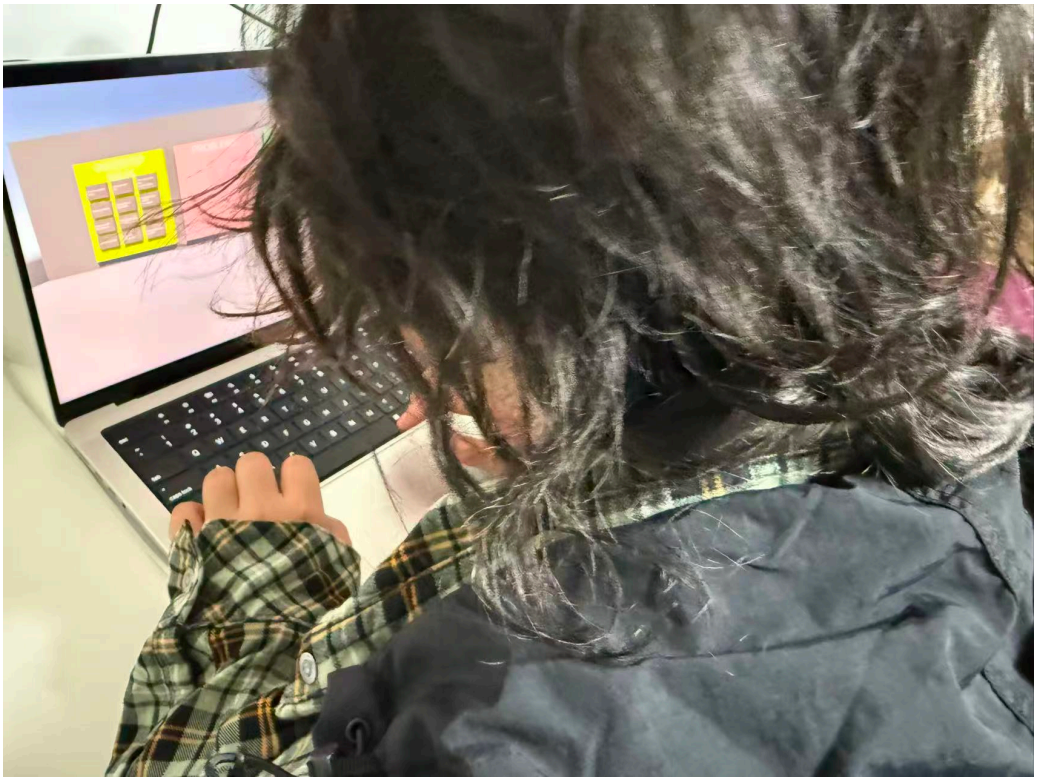
For the next prototype, I will move from the already-supported question “Do users understand spatial organisation?” to a more focused vertical interaction question: “Can direct XR manipulation plus feedforward/feedback make organising multiple ideas immediately discoverable and controllable?” The next test should measure first-action success, mis-drops, time for multi-card organisation and perceived control, while comparing the interaction before and after the added snap/highlight feedback. Collaboration remains part of the wider concept but should only be tested after the single-user manipulation is reliable.

Appendix A – Testing Photos

Participant 2 was the Week 5 studio tutor; a Zoom recording was ongoing, so no photos were taken. Participants 3 and 4 declined to be photographed, resulting in no photo records. During this testing session, photographic records were retained only for the first and fifth participants.



P1



P2

Appendix B - Raw Observation Summary

Participant	Task 1: Understands the spatial zones	Task 2: Move the problem card from Unsorted to Problems	finish time	require help?	error/hesitate	Task 3: Understands the overall purpose	Overall Assessment/5
P1	Correctly Understanding Unsorted and Problems	success	1min23s	no	No obvious errors	Can accurately explain 'organizing and categorizing ideas'.	5
P2	Correctly Understanding Zones	success	2min14s	no	Hesitated slightly the first time I tried to move the object.	Understands that large amounts of information can be organized by category.	4
P3	Got the basic idea, but at first not sure if 'Problems' meant 'actual problems' or 'pending tasks'.	success	2min44s	no	Looked back and forth between the two zones for about 6s.	Can correctly explain the system's purpose.	4
P4	Correctly Understanding Zones	success	1min	no	Brief pause before the first interaction.	Understands the spatial categorization logic.	5
P5	Understood 'Unsorted', but not immediately realize that the cards could be moved.	success	3min15s	yes, 1 gentle nudge.	Unsure how to get started.	Fully understands the system logic after being prompted.	4

Appendix C - Raw Post-Test Responses

P1

P1	
Was it clear what you were supposed to do?	yes
How easy was it to understand where the idea should go?	the processes are clear
Was there anything confusing about the layout or labels?	sometimes confuse about what the ideas mean
What did you expect to happen when you interacted with the idea?	no
What would you change to make this interaction clearer?	maybe it's better to add ueser's hands in it.

P2

P2	
Was it clear what you were supposed to do?	Yes
How easy was it to understand where the idea should go?	It's easy once I decide which category this idea belongs to, I just go to the corresponding board. The most tricky part is to judge which category should they be lol.
Was there anything confusing about the layout or labels?	Not really, maybe my understanding but I felt some cases the "positive finding" also fits as "opportunity", more like a interpretation thing on design knowledge, not something relate to the interactions itself.
What did you expect to happen when you interacted with the idea?	E on notes on board: pick it; Once picked, hold it in the hand; until press E near a board to pin it on the board.
What would you change to make this interaction clearer?	For now it's clear with the instructions. Maybe one thing I will suggest is to add multi selection features, allow us to select multiple notes and hold them together at one time. Now we can only do this one by one, and it's time-consuming. Most time we can decide several notes all belongs to one category and be able to move them together will be nice, more convenient.

P3

P3	
Was it clear what you were supposed to do?	Mostly yes. I understood that I needed to organise the cards, but at first I wasn't completely sure what "Problems" meant.
How easy was it to understand where the idea should go?	It was fairly easy after I understood what each zone represented. I needed a few seconds to compare the zones before deciding.
Was there anything confusing about the layout or labels?	The layout was clear, but the label "Problems" was a little ambiguous to me. I first thought it might mean tasks that still needed to be completed.
What did you expect to happen when you interacted with the idea?	I expected to be able to pick up the card and move it into the appropriate area.
What would you change to make this interaction clearer?	Maybe add a short description or example under each category label so users can understand the meaning of the zones immediately.

P4

P4	
Was it clear what you were supposed to do?	Yes, it was quite clear that I needed to move the card into the correct category.
How easy was it to understand where the idea should go?	Very easy. The different areas made the categorisation clear.
Was there anything confusing about the layout or labels?	Not really. The labels and layout were straightforward.
What did you expect to happen when you interacted with the idea?	I expected the card to be picked up and then placed on the selected board.
What would you change to make this interaction clearer?	Maybe add a small visual highlight when a card can be picked up or when I am close enough to a board to place it.

P5

P5	
Was it clear what you were supposed to do?	I understood that the cards needed to be organised, but at first I wasn't sure how to start interacting with them.
How easy was it to understand where the idea should go?	The categories were understandable, but figuring out how to move the card was more difficult than deciding where it should go.
Was there anything confusing about the layout or labels?	The labels were mostly clear. The confusing part was that I didn't immediately realise the cards were interactive and could be moved.
What did you expect to happen when you interacted with the idea?	At first I expected some instruction or button to appear telling me how to interact with the card. After the hint, I understood that I could pick it up and move it.
What would you change to make this interaction clearer?	Add a visible interaction prompt, such as "Press E to pick up", or highlight the card when the user gets close to it.